LUNA OS

THE AI-NATIVE CLINICAL COMPANION

"Your body, finally understood."

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SLIDE 02 // CORE FOUNDATION

Introduction & Workflow

Luna OS is a high-performance AI companion engineered for women's health. Built specifically to handle complex biometric data and clinical symptoms, it ensures real-time, personalized care for the menopause transition.

Symptom Transition

Adapts seamlessly to tracking user profiles and transitions over multi-decade hormonal windows.

Biometric Ingestion

The platform captures live telemetry metrics like HRV (Heart Rate Variability), skin temperature, and sleep architecture via wearable integration.

Background AI

To guarantee an uninterrupted experience, heavy pattern-recognition (analyzing a 30-day longitudinal history of 34 symptoms) is offloaded to our background LLM engine.

SLIDE 03 // MARKET GAPS

Problem Statement

Complexity

A standard 12-minute doctor visit is completely insufficient to track and diagnose up to 34 distinct, interacting symptoms spanning over decades.

Vulnerability

80% of OB-GYNs lack formal menopause training. Over 1.1 Billion women are affected globally, resulting in $150B lost annually in workplace productivity due to unmanaged symptoms.

Dependency

Existing FemTech platforms are merely menstrual trackers. They act as static lookup tables instead of natively analyzing complex perimenopausal transitions over time.

SLIDE 04 // ARCHITECTURE DESIGN

Solution Approach

Data Privacy (Local-First)

Isolates PII (Personally Identifiable Information) from the AI layer to maintain strict HIPAA-aligned safety protocols and prevent data collisions.

Real-Time Safety Thread

Shifts emergency keyword classification to a prioritized loop. Instantly bypasses the AI and surfaces crisis lifelines if severe medical symptoms are detected.

Context AI Engine

Injects the user's 30-day continuous symptom history directly into the LLM. Analyzes daily metrics against NAMS 2022 clinical guidelines to forecast trigger events natively.

SLIDE 05 // TECHNICAL GUARDRAILS

System Resilience

Pattern Concurrency

Handles overlapping health inputs smoothly. Uses advanced RAG to cross-reference multiple clinical databases (vasomotor, sleep, psychiatry) instantly.

Clinical Stability

Keeps AI outputs strictly evidence-based. Prevents AI hallucinations and medical over-reliance by refusing to recommend specific prescription dosages.

Emergency Persistence

Enforces secure escalation paths. Any input matching crisis keywords triggers a hard-coded safety override.

SLIDE 06 // LIVE METRICS

Clinical Intelligence

Luna OS integrates a native background predictive AI model to monitor, analyze, and forecast clinical transitions continuously.

- Monitors sleep architecture, daily mood logs, and wearables telemetry.
- Automatically builds a real-time, personalized 24-Hour Hot Flash Risk Forecast.
- Generates a proactive "Doctor Prep" clinical dossier without manual input, saving vital physician time.

87%

FORECAST ACCURACY

MODEL: LUNA-TRANSITION-V1

SLIDE 07 // DELIVERY MODEL

Architects, Core Stack & Roadmap

THE ARCHITECTS

Sonu Sharma
Lead Full-Stack & Systems Engineer

Abdisa Jemal Hasen
Lead AI & Prompt Engineer

CORE STACK

Systems Frontend
React Native framework with high-contrast Tech-Noir Tailwind CSS

Intelligence Plane
Anthropic Claude API for intent triage & RAG-grounded response generation

Backend Concurrency
Node.js API with PostgreSQL, running via Replit Agent

PROJECT ROADMAP

P1 Hackathon Baseline
conversational AI & emergency triage safeguards.

P2 Closed Beta
Onboarding 50 real users to validate trigger-detection.

P3 Zenith Scalability
Stripe subscription & telehealth employer pilots.